Prevalence and Associated Factors of Postpartum Depression among Adolescent and Adult Mothers in Khulna region, Bangladesh

Course Title: Masters Level Seminar

Faculty: Professor Dipak Kumar Mitra

Md. Tarek Hasan
(NSU ID: 2517605)

Background

- Postpartum Depression (PPD) prevalence: Globally 17.22%; LMICs 18.6%; Bangladesh 27.75%
- Globally and in LMICs, adolescent mothers have higher PPD prevalence than adults; however, in Bangladesh adult mothers have higher PPD prevalence.
- 30 risk factors and 5 protective factors for PPD were identified among Bangladeshi postpartum mothers.

Justifications of the Study

- Globally and in LMICs, adolescent mothers have higher PPD prevalence than adults; however, in Bangladesh adult mothers have higher PPD prevalence.
- No comparative cross-sectional study was conducted to estimate and compare adolescent and adult PPD prevalence and associated factors in Bangladesh
- Associated factors may vary among adolescent and adult postpartum mothers

Research Questions

Primary Research Question

- What are the prevalence of PPD among adolescent and adult mothers in Khulna region, Bangladesh?

Secondary Research Question

- What are the socio-demographic and obstetric factors associated with postpartum depression among adolescent and adult mothers in Khulna region, Bangladesh?

Objectives

General Objective

- To assess the prevalence and associated factors of PPD among adolescent and adult mothers in Khulna region, Bangladesh

Specific Objectives

- To estimate the prevalence of PPD among adolescent and adult mothers
- To measure the association between participants socio-demographic characteristics and PPD
- To measure the association between participants obstetric characteristics and PPD

Conceptual Framework

Independent Variables

- Sociodemographic Variables
  - Maternal Age
  - Education Level
  - Maternal Occupation
  - Household Income
  - Family Type
  - Residence
- Obstetric Variables
  - Parity
  - Pregnancy Wantedness
  - ANC Visits
  - Neonate Gender
  - Last Delivery Method

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Dependent Variable

Postpartum Depression (EPDS Score $\geq$10)

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Adolescent Mothers

Adult Mothers

Research Methodology

- **Study Design:** Comparative cross-sectional study
- **Study Population:** Adolescent and adult postpartum mothers in Khulna region
- **Study Site:** Khulna Medical College and Gazi Medical College
- **Study Duration:** Between February 2026 and September 2026

Research Methodology (cont.)

- **Inclusion Criteria:**
  - Adolescent mothers (age 15-19 years) within 1 year postpartum
  - Adult mothers (age 20-49 years) within 1 year postpartum
- **Exclusion Criteria:**
  - Mothers having newborns with severe illness
  - Mothers having chronic illness
  - Mothers having preexisting mental health disorder

Sample Size Calculation

$$n = \frac{r+1}{r} \cdot \frac{\bar{p}(1-\bar{p})(Z_\beta + Z_{\alpha/2})^2}{(p_1 - p_2)^2}$$
$$= \frac{(1+1)}{1} \cdot \frac{0.181(1-0.181)(1.96+0.84)^2}{(0.118-0.243)^2}$$
$$= 148 \text{ per group}$$

Assumptions

- $p_1 = 0.118$ (Teen)
- $p_2 = 0.243$ (Adult)
- $\bar{p} = \frac{0.118+0.243}{2} = 0.181$
- $Z_{\alpha/2} = 1.96$
- $Z_\beta = 0.84$
- $r = 1$

$$\text{Total} = (148 + 148)$$
$$= \mathbf{296}$$

Research Methodology (cont.)

Sampling Technique:

- A **convenience sampling** approach will be used.

Data Collection Tools:

- Bengali version of **Edinburgh Postnatal Depression Scale** (EPDS-B) (10 items)
- **Socio-demographic** questionnaire
- **Obstetric** questionnaire

Data Analysis Plan

Required Softwares:

- MS excelsheet for data entry; R programming for data analysis

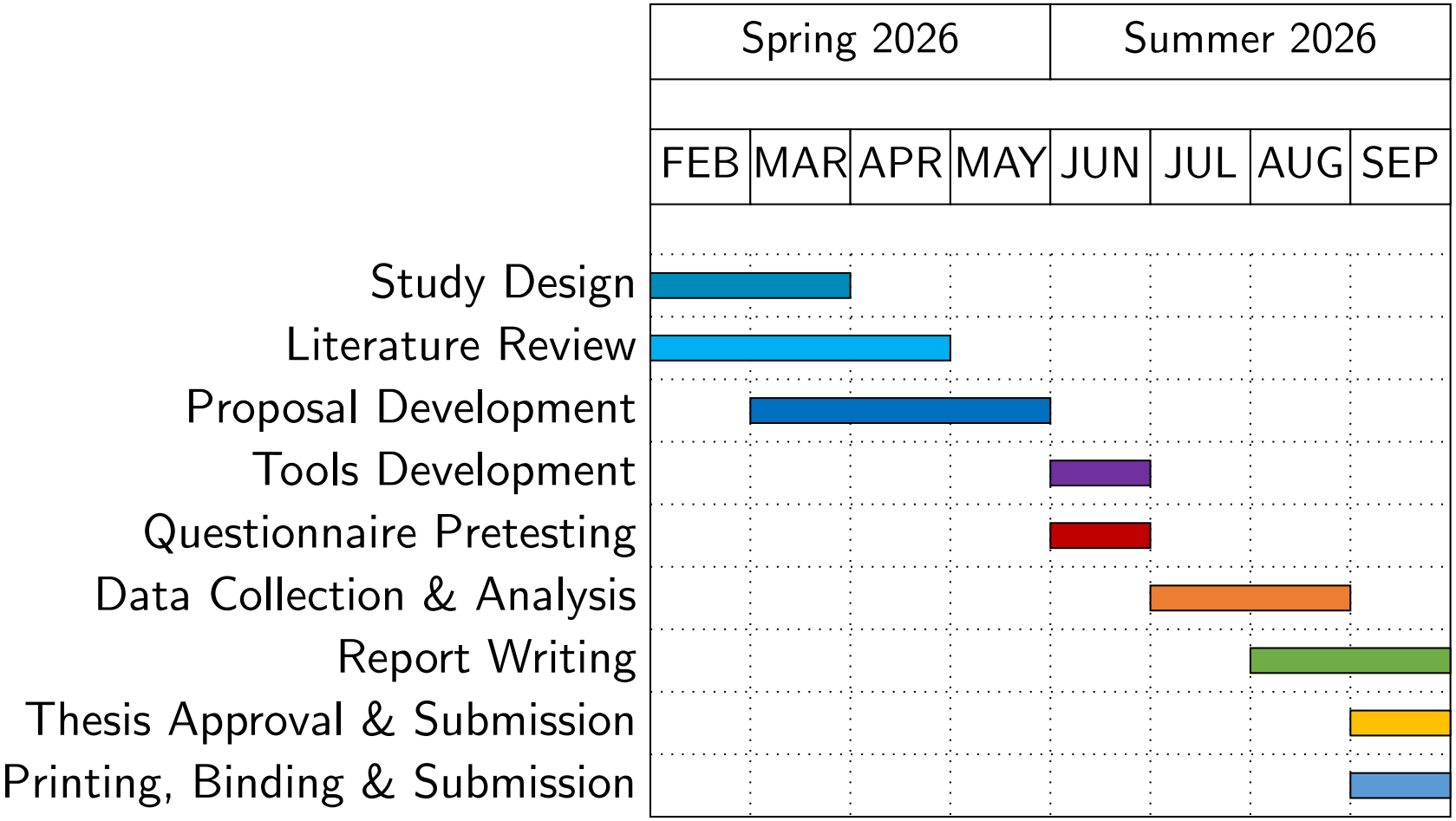
Required Analysis:

- **Descriptive statistics** to describe PPD prevalence and socio-demographic characteristics
- χ^2 **test** to compare categorical data between adolescent and adult mothers
- **Bivariate logistic regression** to explore the associations between PPD and each of the selected explanatory variables separately for adolescent and adult mothers
- **Multivariable logistic regression** to measure adjusted association between PPD and explanatory variables in the two groups

Research Methodology (cont.)

- Prior to data collection, ethical approval will be obtained from the **Institutional Review Board (IRB) of North South University.**

Study Timeline



Thank You